



RIVER VALLEY HIGH SCHOOL

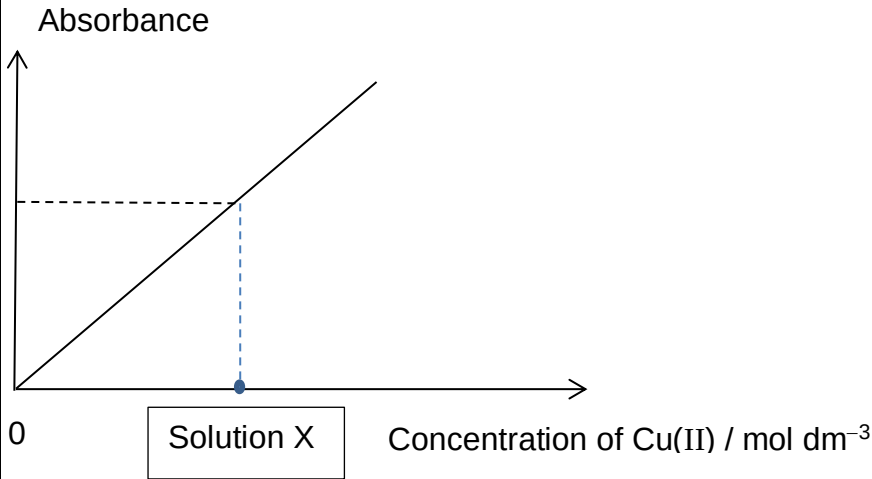
H2 CHEMISTRY

Suggested Solutions for 2013 A Level H2 Chemistry Paper 1

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	C
2	B	22	D
3	D	23	B
4	C	24	B
5	A	25	B
6	C	26	B
7	A	27	C
8	B	28	C
9	D	29	A
10	C	30	D
11	D	31	A
12	B	32	D
13	D	33	C
14	D	34	B
15	B	35	D
16	D	36	B
17	D	37	A
18	A	38	B
19	C	39	C
20	C	40	A

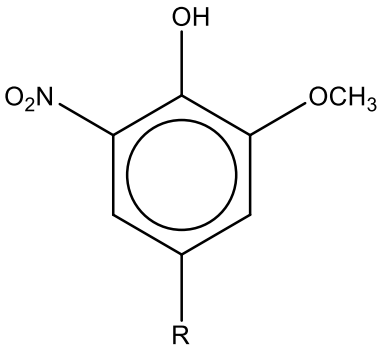
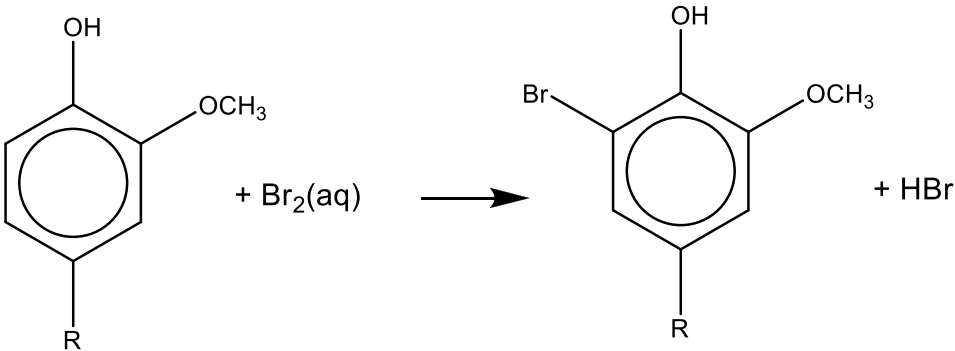
Suggested Solutions for 2013 A Level H2 Chemistry Paper 2

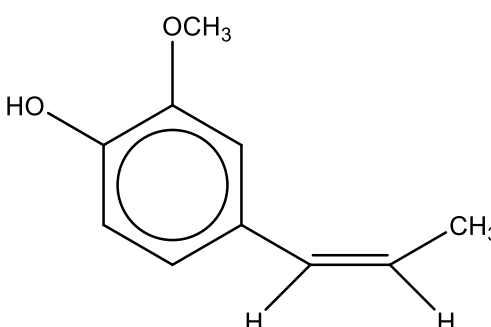
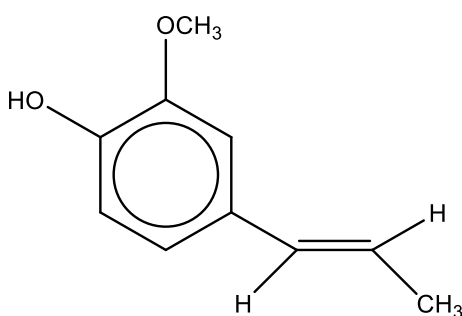
1	(a)	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	[1]										
	(b)	Colour: orange Explanation: The colour absorbed should be the complement of blue.	[1]										
	(c)	<u>Preparation of 250.0 cm³ of 2.00 mol dm⁻³ CuSO₄(aq)</u> Mass of CuSO₄•5H₂O(s) required = $\frac{250}{1000} \times 2.00 \times 249.6$ = 124.8 g = 125 g Weigh accurately a clean dry empty 100 cm³ beaker using a weighing balance and record the mass. Place about 125 g of CuSO₄•5H₂O(s) in the beaker and reweigh. Calculate the accurate mass of the CuSO₄•5H₂O(s) by subtracting the mass of the empty beaker from the mass of beaker and CuSO₄•5H₂O(s). Dissolve the CuSO₄•5H₂O(s) in deionised water. Transfer quantitatively the CuSO₄ solution and rinsings to a 250.0 cm³ volumetric flask. Make-up to the mark with deionised water adding dropwise close to mark and shake the flask well to obtain a homogenous solution. <u>Preparation of diluted solutions</u> Volume of 2.00 mol dm⁻³ CuSO₄(aq) to be used = (desired concentration of CuSO₄(aq) × 100.0) / 2.00 The following table shows the required volume of 2.00 mol dm⁻³ CuSO₄(aq) to be used to prepare 100 cm³ of CuSO₄(aq) of the desired concentration. <table><tr><th>Concentration of CuSO₄(aq) / mol dm⁻³</th><th>Volume of 2.00 mol dm⁻³ CuSO₄(aq) / cm³</th></tr><tr><td>1.60</td><td>80.0</td></tr><tr><td>1.20</td><td>60.0</td></tr><tr><td>0.80</td><td>40.0</td></tr><tr><td>0.40</td><td>20.0</td></tr></table>	Concentration of CuSO ₄ (aq) / mol dm ⁻³	Volume of 2.00 mol dm ⁻³ CuSO ₄ (aq) / cm ³	1.60	80.0	1.20	60.0	0.80	40.0	0.40	20.0	
Concentration of CuSO ₄ (aq) / mol dm ⁻³	Volume of 2.00 mol dm ⁻³ CuSO ₄ (aq) / cm ³												
1.60	80.0												
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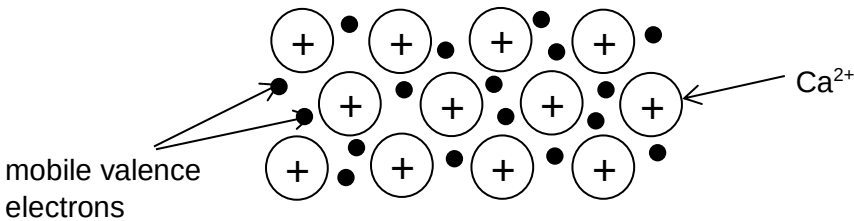
	To prepare the diluted $\text{CuSO}_4(\text{aq})$ solutions, the appropriate volume of $2.00 \text{ mol dm}^{-3} \text{ CuSO}_4(\text{aq})$ as shown in the table above is transferred using a burette to a 100 cm^3 volumetric flask and topped up to the mark with deionised water. The flask is then well shaken to produce a homogeneous solution.	
	<u>Preparation of Cu complex</u> 10.0 cm^3 of the $\text{CuSO}_4(\text{aq})$ solutions are separately pipetted into separate 100 cm^3 volumetric flask and topped up to the mark with $2.00 \text{ mol dm}^{-3} \text{ NH}_3(\text{aq})$. 10.0 cm^3 solution X is also separately pipetted into a separate 100 cm^3 volumetric flask and topped up to the mark with $2.00 \text{ mol dm}^{-3} \text{ NH}_3(\text{aq})$. (The volume of $\text{NH}_3(\text{aq})$ added ensures a large excess of $\text{NH}_3(\text{aq})$ since the minimum volume of $\text{NH}_3(\text{aq})$ required is 40 cm^3.) The absorbance of the Cu complex solutions is measured. A graph of absorbance against concentration of the $\text{CuSO}_4(\text{aq})$ used is plotted. A straight line passing through the origin should be obtained.  The absorbance given by the Cu complex solution prepared from solution X is measured and the corresponding concentration read off from the calibration graph. (Note: Candidates then had to describe a workable method to prepare at least five reference solutions of different concentration from the 250.0 cm^3 of standard solution that were well spread between 0 and 2.00 mol dm^{-3}. They could use any suitable combination of pipettes, burettes and/or volumetric flasks. Some used less precise apparatus, such as measuring cylinders or did not specify the apparatus at all.)	[10]
	[Total: 12]	

2	(a)	(i)	Reaction (1) is responsible as it is the most exothermic.	
		(ii)	There is a net increase in the number of gas molecules for reactions (2) and (4), whereas there is no net change in number of gas molecules for the other reactions. (Note: do not give vague answers such as "more gaseous products")	
		(iii)	$\Delta G = \Delta H - T\Delta S$ $= (-24.8 \times 10^3) - [(773)(+15.8)]$ $= -37013 \text{ J mol}^{-1}$ $= -37.0 \text{ kJ mol}^{-1}$ Since ΔS is positive, $-T\Delta S$ is negative. Increasing the temperature makes $-T\Delta S$ more negative. Hence ΔG becoming more negative and the reaction is more spontaneous.	
		(iv)	<u>Le Chatelier's Principle states that if a system at equilibrium is subjected to a change which disturbs the equilibrium, the system responds in such a way to counteract the effect of the change.</u> Increasing temperature favours the forward endothermic reaction to absorb some of the added heat. Hence the equilibrium position of reaction (2) will shift to the right, increasing the partial pressure of CO. Increasing pressure would cause a decrease in the partial pressure of carbon monoxide. The position of equilibrium of reaction (2) would shift left to relieve some of the increased pressure by reducing the number of gaseous molecules.	
		(v)	Amount of SiO_2 present = $(\frac{3.5}{100} \times 1 \times 10^6) \div 60.1 = 582 \text{ mol}$ $\text{SiO}_2 \equiv \text{CaO} \equiv \text{CaCO}_3$ Mass of $\text{CaCO}_3 = 582 \times [(40.1 + 12.0 + 3(16.0))] = 5.82 \times 10^4 \text{ g}$ $= 58.2 \text{ kg}$	[9]
	(b)	(i)	$2\text{Fe}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{Sn}^{4+}(\text{aq})$	
		(ii)	$E^\ominus_{\text{cell}} = +0.77 - (+0.15) = +0.62\text{V}$	
		(iii)	$\text{Fe}^{2+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow \text{Fe}(\text{s}) + \text{Sn}^{4+}(\text{aq})$ $E^\ominus_{\text{cell}} = -0.44 - (+0.15) = -0.59\text{V} < 0$ Hence the reaction is non-spontaneous. For the reduction of Fe^{3+} to Fe^{2+} by Sn^{2+}, $E^\ominus_{\text{cell}}$ is positive. Hence the reaction is spontaneous.	[4]

	(c)	(i)	Titration No. 1 is excluded. Average titre = 21.45 cm ³ $\text{Amount of K}_2\text{Cr}_2\text{O}_7 = \frac{21.45}{1000} \times 0.100$ $= 2.15 \times 10^{-3} \text{ mol}$	
		(ii)	$6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ \rightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	
		(iii)	Amount of Fe ²⁺ in original solution = $(6 \times 2.15 \times 10^{-3} \times 250) \div 25.0$ $= 0.129 \text{ mol}$ % by mass = $[(0.129 \times 55.8) \div 11.15] \times 100\%$ $= 64.6\%$ (May obtain 64.4% if calculate using more s.f)	[5]
			[Total: 18]	

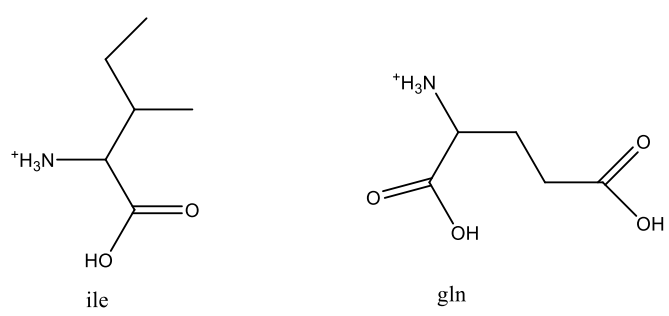
3	(a)	(i)	Reagent A : NaOH(aq) Reagent B : CH(CH ₃) ₂ COCl																					
		(ii)																						
		(iii)	Orange aqueous bromine is decolourised. White precipitate formed.																					
		(iv)		[5]																				
	(b)	(i)	$HA \rightleftharpoons H^+ + A^-$ A more stable anion results in the position of equilibrium shifting more to the right and a stronger acid. The <u>p orbital of oxygen overlaps with the π electron cloud of the benzene ring</u> . The <u>negative charge on oxygen is delocalised</u> into the ring, thereby <u>stabilising the phenoxide ion</u> . Ethyl group is electron-donating and <u>intensifies the negative charge on oxygen</u> , thereby <u>destabilising the ethoxide ion</u> . Thus phenol is a stronger acid compared to ethanol.																					
		(ii)	<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 15%;"></td> <td style="width: 15%; text-align: center;">HA</td> <td style="width: 10%; text-align: center;">$\rightleftharpoons$</td> <td style="width: 15%; text-align: center;">$A^-(aq) +$</td> <td style="width: 15%; text-align: center;">$H^+(aq)$</td> </tr> <tr> <td>Initial conc / mol dm⁻³</td> <td style="text-align: center;">0.001</td> <td></td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> </tr> <tr> <td>Change / mol dm⁻³</td> <td style="text-align: center;">-x</td> <td></td> <td style="text-align: center;">+x</td> <td style="text-align: center;">+x</td> </tr> <tr> <td>Eqm conc/ mol dm⁻³</td> <td style="text-align: center;">0.001-x</td> <td></td> <td style="text-align: center;">x</td> <td style="text-align: center;">x</td> </tr> </table>		HA	$\rightleftharpoons$	$A^-(aq) +$	$H^+(aq)$	Initial conc / mol dm ⁻³	0.001		0	0	Change / mol dm ⁻³	-x		+x	+x	Eqm conc/ mol dm ⁻³	0.001-x		x	x	
	HA	$\rightleftharpoons$	$A^-(aq) +$	$H^+(aq)$																				
Initial conc / mol dm ⁻³	0.001		0	0																				
Change / mol dm ⁻³	-x		+x	+x																				
Eqm conc/ mol dm ⁻³	0.001-x		x	x																				

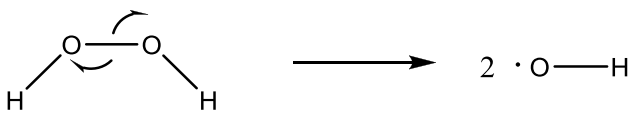
		$K_a = \frac{(x)(x)}{0.001 - x}$ Assume x is very small such that $0.001 - x \approx 0.001$, $K_a = 10^{-7.4} \approx \frac{(x)^2}{0.001}$ $[H^+] = x = 6.31 \times 10^{-6} \text{ mol dm}^{-3}$ $\text{pH} = -\lg(6.31 \times 10^{-6})$ $= 5.20$ OR $[H^+] = \sqrt{0.001 \times 10^{-7.4}}$ $= 6.31 \times 10^{-6} \text{ mol dm}^{-3}$ $\text{pH} = -\lg(6.31 \times 10^{-6})$ $= 5.20$	[5]
(c)	cis isomer 	trans isomer 	[2]
			[Total: 12]

4	(a)	(i)	Nuclear charge is higher for iron and copper than calcium, while shielding effect by inner shells of electrons remains relatively constant as 3d electrons provide relatively poor screening effect. Hence, effective nuclear charge is higher for iron and copper than for calcium. Attraction between the valence electrons and nucleus is stronger in iron and copper than calcium, resulting in smaller atomic radii for iron and copper. (Note: quite a number only explained the difference in terms of no. of protons and shielding effect. Did not go on to explain how this will affect the attraction for outer electrons and hence the radius.)	
		(ii)	The relative atomic mass of iron (55.8) and copper (63.5) are higher than that of calcium (40.1). The atomic radii of iron (0.126 nm) and copper (0.128 nm) are smaller than that of calcium (0.197 nm). Since density is mass per unit volume, with a higher relative atomic mass and smaller radius, the iron and copper atoms are packed more closely and the density is higher.	[4]
	(b)	(i)	Calcium has a <u>giant metallic structure</u>. <u>Electrostatic forces of attractions exist between the Ca^{2+} ions and the sea of mobile delocalised valence electrons.</u>  (Note: question required the structure as well as bonding. Some candidates chose to use vague terms, describing electrons as 'free electrons' or sea of electrons'.)	
		(ii)	The ionic radius of Fe^{2+} is smaller than Ca^{2+}. The strength of the electrostatic forces of attractions between Fe^{2+} and the delocalised electrons is stronger than that between the Ca^{2+} and the delocalised electrons. Therefore, the melting point of iron is higher than that of calcium.	[3]
	(c)	(i)	Copper wires will be <u>very heavy</u> (due to the high density of copper). (Note: The overhead wires might sag and strong reinforcements will be required to prevent sagging.)	
		(ii)	Cu $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ Cu^{2+} $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$	

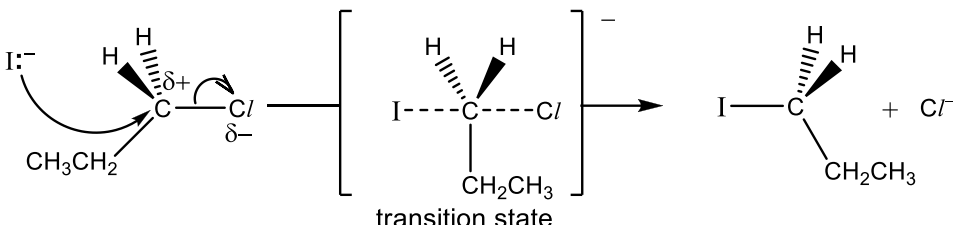
		(iii)	Copper is purified using <u>electrolysis</u>, with <u>CuSO₄(aq)</u> as the <u>electrolyte</u>. The anode is the impure copper metal, while a piece of pure copper metal acts as the cathode. <u>Cu is oxidised to Cu²⁺ at the anode, while Cu²⁺ is reduced to Cu at the cathode.</u> (Note: Do not give excessive descriptions of redox reactions involving other impurity metals – “Outline” was used in the question.)	[5]
	(d)	(i)	(Not required in 9729 syllabus. Need to know decomposition of carbonates instead) $\text{Ca(NO}_3)_2 \xrightarrow{\text{D}} \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	
		(ii)	Cu(NO₃)₂ will decompose at a <u>lower</u> temperature. Cu²⁺ (0.069 nm) has a <u>smaller ionic radius</u> than Ca²⁺ (0.099 nm) and hence <u>charge density and polarising power of Cu²⁺ is greater</u> than Ca²⁺. NO₃⁻ anions are polarised to a greater extent by Cu²⁺, and the <u>N–O bonds in Cu(NO₃)₂ are weakened to a greater extent.</u> Hence Cu(NO₃)₂ is less thermally stable.	[3]
			[Total: 15]	

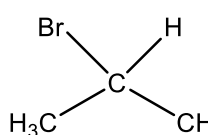
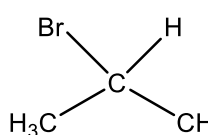
(For Q5, Only part (e) and (f) are in the 9729 syllabus)

5	(a)	Support / Transport / Defence / Enzyme (Any 3)		[1]
	(b)	(i)	cys–tyr–ile–gln–asn–cys–pro–leu–gly (Note: Start from N terminal and end at C terminal)	
		(ii)	Phenol group on the side chain of tyrosine residue (Note: The amide NH ₂ groups are neutral and cannot be ionised.)	[2]
	(c)	Tertiary structure Disulfide bridges can be formed between the –SH groups of two cysteine. Van der Waals' forces of attraction can be formed between the side chains of leucine, isoleucine, tyrosine and proline. Hydrogen bonds can be formed between the oxygen lone pair on the –OH of tyrosine and the protonic hydrogen on the –CONH₂ of asparagine and glutamine. Ionic interactions can be formed between the –O[–] of tyrosine and the –NH₃⁺ of the N terminal amine group of cysteine. (Note: Answer should make reference to the specific amino acids in oxytocin and be precise enough about which groups are involved. Interactions between other pairs of amino acids are possible. Diagrams can be used to better illustrate the R group interactions.)		[4]
	(d)	Hg²⁺ ions can <u>form ionic interactions with –COO[–]</u>, which brings about the formation of insoluble protein salts (i.e. precipitation of protein). This disrupts the <u>ionic interactions</u> in the tertiary structure, leading to denaturation. Heavy metal ions also have a high affinity for sulfur and will <u>bind tightly to the –SH group</u> of the cysteine residues. This disrupts the <u>disulfide bridges</u> in the tertiary structure, leading to denaturation.		[2]
	(e)	(i)	Complete acidic hydrolysis of protein	
		(ii)	 (Note: This hydrolysis took place in acidic conditions. Some	[4]

			candidates missed out on the protonation of amine groups, hydrolysis of amide side chain on glutamine (gln). Some incorrectly protonated amide nitrogen of glutamine.)	
	(f)	 A free radical is an atom or a group of atoms with one <u>unpaired valence electron</u>. •OH free radicals are formed when O-O covalent bond undergoes homolytic fission. (Note: Fewer candidates correctly defined a free radical.)		[2]
			[Total: 15]	

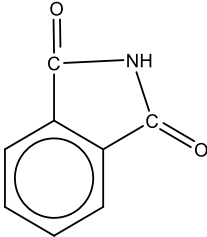
Suggested Solutions for 2013 A Level H2 Chemistry Paper 3

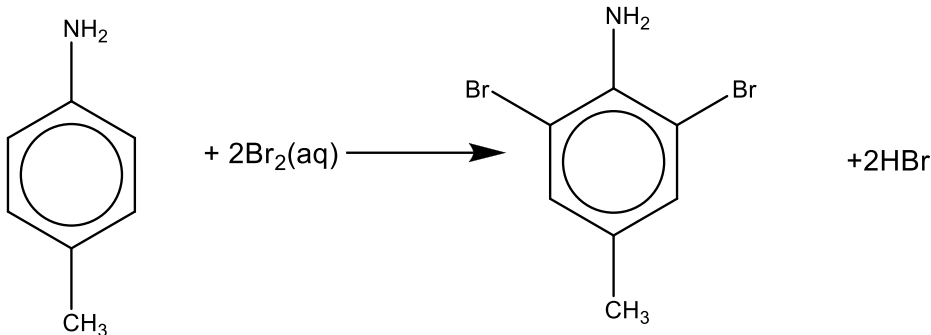
1	(a)	<table><thead><tr><th>Element</th><th>$E^{\ominus} / \text{V}$</th></tr></thead><tbody><tr><td>$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$</td><td>+2.87</td></tr><tr><td>$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$</td><td>+1.36</td></tr><tr><td>$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$</td><td>+1.07</td></tr><tr><td>$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$</td><td>+0.54</td></tr></tbody></table> Oxidising power of halogens decreases down the group. Down the group, $E^{\ominus}(\text{X}_2/\text{X}^-)$ becomes less positive and the halogen is less readily reduced. (Note: Wrong phrasing that is seen from our centre: “as the $E^{\ominus}$ values decrease down the group, the reducing power also decreases, so the oxidising power increases down the group”.)	Element	$E^{\ominus} / \text{V}$	$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87	$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36	$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.07	$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+0.54	[2]
Element	$E^{\ominus} / \text{V}$												
$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87												
$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36												
$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.07												
$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+0.54												
	(b)	(i) $2\text{HX} \rightarrow \text{H}_2 + \text{X}_2$ (Note: Wrong answers: $\text{HX} \rightarrow \text{H}\cdot + \text{X}\cdot$ or $\text{HX} \rightarrow \text{H}^+ + \text{X}^-$.)											
		(ii) The thermal stability of hydrogen halides decreases down the group due to decreasing H–X bond energy. The size of the halogen atoms increases from F to I and their valence orbitals become more diffuse, resulting in less effective overlap of the orbitals between the H atom and halogen atom and less energy is required to break the weaker H–X bond. (Note: Some candidates wrote a significant amount of irrelevant, factually correct material on bond lengths, electronegativities and ionisation energies. A number of candidates also wrote about intermolecular forces, clearly confusing thermal stability with volatility.)											
	(c)	(i)  (Note: A significant number of candidates incorrectly drew the attacking species as a covalent Na-I molecule, when it is obviously an ionic compound. Some omitted the lone pair on iodide ion. Some candidates did not show the charge on the transition state of the $\text{S}_{\text{N}}2$ reaction and all relevant δ^+/δ^- partial charges.)											

		(ii)	The formation of insoluble sodium chloride precipitate in propanone will shift the equilibrium position of equilibrium 1 to the right and drive the forward reaction.																					
		(iii)	Let the M_r of the alkyl portion of the A and D be R Since the M_r of D is 38.2% larger than the M_r of A $(R + 79.9)\left(\frac{138.2}{100}\right) = (R + 127)$ $R = 43.3$ Since the alkyl portion of A has the formula C_nH_{2n+1}, $12n + 1(2n+1) = 43.3$ $n = 3$ The alkyl portion has the formula C_3H_7. So A is C_3H_7Br. A undergoes nucleophilic substitution to form B, an alcohol. Since B can be oxidised to form a neutral compound C, C must be a ketone. Hence, B must be a secondary alcohol and A is a secondary alkylbromide.  The structure of A is H_3C  CH_3.																					
	(d)	(i)	$K_c = \frac{[I_3^-]}{[I_2][I^-]}$; units: $\text{mol}^{-1} \text{dm}^3$																					
		(ii)	Initial concentration of $I_2(\text{aq}) = \left(\frac{2.54}{127 \times 2}\right) \div \left(\frac{100}{1000}\right)$ $= 0.100 \text{ mol dm}^{-3}$ <table border="1" style="margin-left: auto; margin-right: auto;"><thead><tr><th></th><th>$I_2(\text{aq})$</th><th>$I^-(\text{aq})$</th><th>$\rightleftharpoons$</th><th>$I_3^-(\text{aq})$</th></tr></thead><tbody><tr><td>I/ mol dm^{-3}</td><td>0.100</td><td>1.00</td><td></td><td>0</td></tr><tr><td>C/ mol dm^{-3}</td><td>-9.98×10^{-2}</td><td>-9.98×10^{-2}</td><td></td><td>$+9.98 \times 10^{-2}$</td></tr><tr><td>E/ mol dm^{-3}</td><td>2.00×10^{-4}</td><td>0.9002</td><td></td><td>9.98×10^{-2}</td></tr></tbody></table> $[I_2(\text{aq})] = 2.00 \times 10^{-4} \text{ mol dm}^{-3}$ $[I^-(\text{aq})] = 0.900 \text{ mol dm}^{-3}$		$I_2(\text{aq})$	$I^-(\text{aq})$	$\rightleftharpoons$	$I_3^-(\text{aq})$	I/ mol dm^{-3}	0.100	1.00		0	C/ mol dm^{-3}	-9.98×10^{-2}	-9.98×10^{-2}		$+9.98 \times 10^{-2}$	E/ mol dm^{-3}	2.00×10^{-4}	0.9002		9.98×10^{-2}	
	$I_2(\text{aq})$	$I^-(\text{aq})$	$\rightleftharpoons$	$I_3^-(\text{aq})$																				
I/ mol dm^{-3}	0.100	1.00		0																				
C/ mol dm^{-3}	-9.98×10^{-2}	-9.98×10^{-2}		$+9.98 \times 10^{-2}$																				
E/ mol dm^{-3}	2.00×10^{-4}	0.9002		9.98×10^{-2}																				
			$K_c = \frac{9.98 \times 10^{-2}}{2.00 \times 10^{-4} \times 0.900} = 554 \text{ mol}^{-1} \text{dm}^3$																					

		(iii)	$K_c = \frac{[I_3^-]}{[I_2][I^-]}$ $554 = \frac{1.00}{[I_2](1.00)}$ $[I_2] = 1.80 \times 10^{-3} \text{ mol dm}^{-3}$	
		(iv)	As hexane is added, more I_2 will dissolve in hexane, thus $[I_2(aq)]$ will decrease. As $[I_2(aq)]$ decreases, according to Le Chatelier's Principle, this will shift the equilibrium position of equilibrium 2 to the left. As a result, $[I^-(aq)]$ will increase and $[I_3^-(aq)]$ will decrease. (Note: Some candidates incorrectly stated that $[I_2(aq)]$ will increase to make up for the iodine that had dissolved in hexane.)	
		(v)	$E^\ominus_{\text{cell}} = +0.536 - (+0.34)$ $= +0.196V$ The I_3^-/I^- half-cell is the reduction half-cell and the Cu^{2+}/Cu half-cell is the oxidation half-cell. From d(iv), the concentration of I_3^- will decrease and the concentration of I^- will increase. $I_3^-(aq) + 2e^- \rightleftharpoons 3I^-(aq)$ For the equilibrium of the reduction half-cell, the decrease in I_3^- concentration and increase in I^- concentration will cause the backward reaction to be favoured (according to Le Chatelier's Principle). This will cause the $E(I_3^-/I^-)$ to be less positive. As a result, the cell potential will be <u>less positive</u> (lower than +0.196V). (Note: Candidates have to consider 2 equilibria. It is not always clear from the answers which of the two equilibria candidates were referring to. Common incorrect answer "Adding hexane will do nothing to the equilibrium position since it does not react with I^- or I_3^-, hence the cell potential remains constant." Some candidates also incorrectly suggested there would be no change in cell potential as it will be determined by the ease of I_3^- reduction rather than $[I_3^-]$.)	[9]
			[Total: 20]	

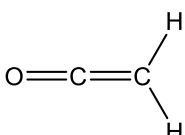
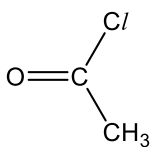
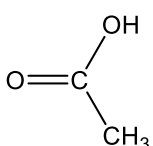
2	(a)	A Bronsted-Lowry base is a substance that can accept a proton from another substance. $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^- \qquad \text{B} + \text{H}_2\text{O} \rightleftharpoons \text{BH}^+ + \text{OH}^-$ A conjugate acid-base pair is an acid and a base differing by the presence or absence of a proton, such as HA & A⁻ and B & BH⁺. (Note: There were a number of misconceptions such as defining the Bronsted-Lowry base as one that gives protons in a reaction, as an electron donor or one that give OH⁻ ions when dissolved in water. Only a few candidates were able to describe a conjugate acid-base pair correctly.)	[2]
	(b)	(i) NH₃ acts as a base. NH₃ accepts a H⁺ to form NH₄⁺ which is a conjugate acid.	
		(ii) NH₃ acts as a reducing agent. In the reaction, oxidation number of N in NH₃ increases from -3 to 0 in N₂ while that of N in HNO₂ decreases from +3 to 0 in N₂. Hence NH₃ is oxidised, thus it acts as a reducing agent.	
		(iii) NH₃ acts as an acid. In the reaction, NH₃ donates a H⁺ to H⁻ ion in NaH to form NH₂⁻ ion and H₂.	
		(iv) NH₃ acts as a reducing agent OCI⁻ is reduced to Cl⁻ where oxidation number of Cl decreases from +1 to -1. Oxidation number of N also increases from -3 in NH₃ to -2 in N₂H₄.	

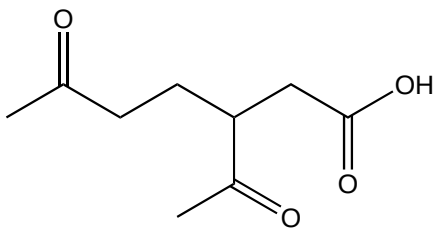
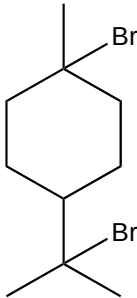
		(v) NH_3 acts as a nucleophile. NH_3 nucleophile donates an electron pair to the acyl carbon in  COCl to form (Note: Explanations were lacking in detail. A number of candidates confused oxidation with reduction and acid with base.)	[5]
	(c)	Ethylamine is most basic followed by ammonia and the least basic compound is phenylamine. Ethylamine is more basic than ammonia because <u>electron-donating ethyl group</u> makes the <u>lone pair of electrons on N atom</u> of ethylamine <u>more available for protonation</u> as compared to ammonia. Comparing phenylamine and ammonia, the <u>2p orbital on N atom</u> of phenylamine is <u>overlaps with the π electron cloud</u> and the <u>lone pair of electrons can delocalise into the benzene ring</u>, making it <u>less available for protonation</u>. Hence, phenylamine is less basic than ammonia. (Note: There were some misconceptions that benzene ring in phenylamine is electron-rich, making the lone pair on nitrogen more available for protonation. Some explanations of the poor basicity of phenylamine were not clear.)	[3]
	(d) (i)	4-methylphenylamine is more basic than phenylamine. This is because there is an <u>electron donating methyl group</u> (which will increase the electron density in the benzene ring). This will <u>decrease the delocalisation of the lone pair of electrons on N atom of $-\text{NH}_2$ group into the benzene ring</u> and thus <u>cause the lone pair of electron on N atom in 4-methylphenylamine to be more available for protonation</u>. Ethylamine is <u>more basic</u> than benzylamine. <u>$\text{C}_6\text{H}_5\text{CH}_2-$ group has a smaller electron donating effect as compared to ethyl group</u>. Hence the <u>lone pair of electrons on N atom of benzylamine is less available for protonation</u>. (Note: Some candidates did not read the question carefully and compared the basicities of 4-methylphenylamine and benzylamine. In the 2nd comparison, the most common error was to mention the weaker basicity of benzylamine as being due to the delocalisation of the lone pair on N over benzene ring. Since the $-\text{NH}_2$ group is separated from the ring by a $-\text{CH}_2$ group, no delocalisation can occur.)	

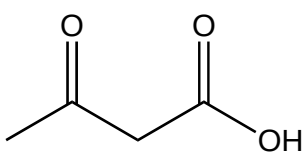
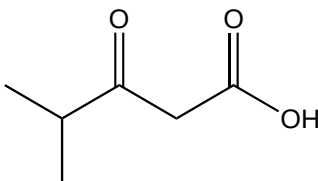
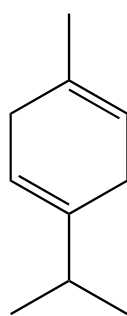
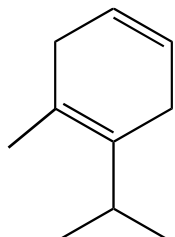
		(ii) Reagent: Br₂ (aq) Orange aqueous bromine decolourised when it is added to 4-methylphenylamine and a white ppt is formed. Orange solution remains when it is added to benzylamine.  (Note: Several candidates thought that benzylamine would react with bromine water but not 4-methylphenylamine. Some suggested that bromine would react with both amine but that only one would give a white ppt. They suggested that bromine would react with the methyl group to give a bromomethyl benzene. They also suggested that bromine would react with benzylamine (replacing the –NH₂ group by Br) but not methylphenylamine.)	
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		(iii) <chem>CC1=CC=CC=C1</chem> $\xrightarrow[\text{maintain at } 30^{\circ}\text{C}]{\text{Conc HNO}_3, \text{Conc H}_2\text{SO}_4}$ <chem>CC1=CC=C([N+](=O)[O-])C=C1</chem> $\xrightarrow[2. \text{NaOH(aq)}]{1. \text{Sn, Conc HCl, heat}}$ <chem>CC1=CC=C(N)C=C1</chem> <chem>CC1=CC=CC=C1</chem> $\xrightarrow[\text{u.v light}]{\text{Br}_2}$ <chem>CC1=CC=CC=C1CBr</chem> $\xrightarrow[\text{Heat in a sealed tube}]{\text{Ethanolic NH}_3}$ <chem>CC1=CC=CC=C1CN</chem> (Note: Answers were often incomplete or incorrect. Several routes to 4-methylphenylamine used aryl chlorides or aryl bromides as intermediates, reacting these with ethanolic ammonia. The intermediate aryl chlorides were variously made by reacting methylbenzene with FeCl_3 (but no Cl_2), PCl_5 or $\text{Cl}_2(\text{aq})$. Those who did use the nitration-reduction route sometimes wrongly used dilute or $\text{HNO}_3(\text{aq})$, or used $\text{Sn} + \text{NaOH}$ as the reducing agent. Errors included using dilute or aqueous, acids for the nitration, reducing the NO_2 group with LiAlH_4 and heating the alcoholic NH_3 under reflux for the second step of route 2. Some candidates wrongly specified that the Cl_2 used in the first step of route 2 was (aq).)	[10]
		[Total: 20]	

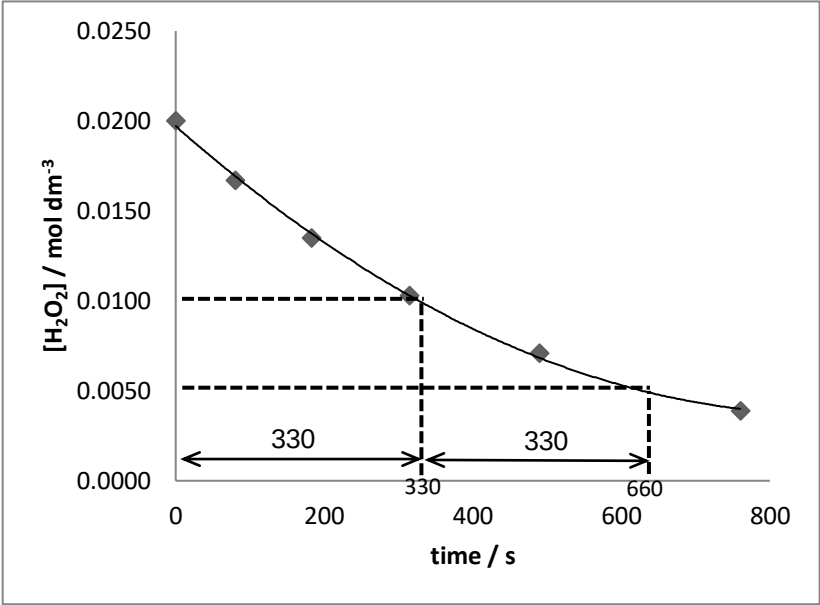
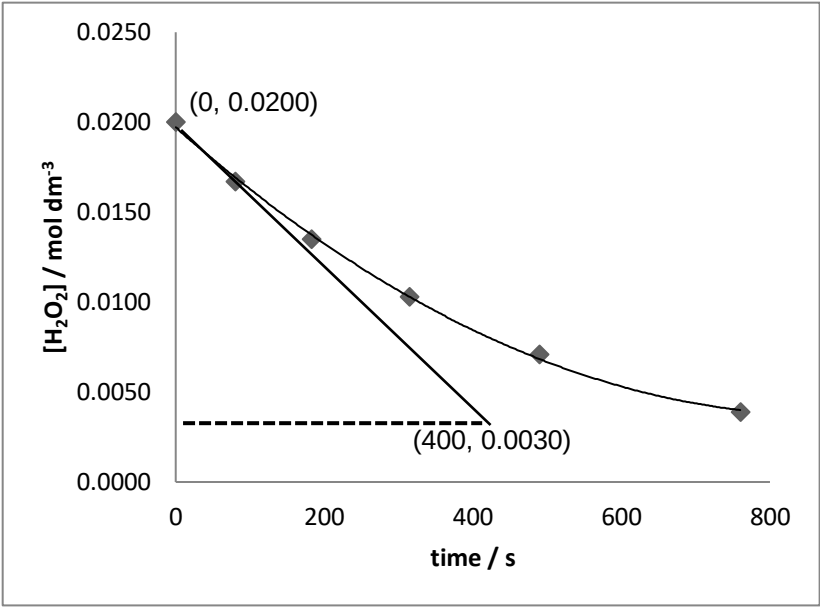
3	(a)	(Reaction of P_4O_6 not in 9729 syllabus) P_4O_6 reacts <u>violently/vigorously</u> with water to give an acidic solution of pH = 2. $P_4O_6(s) + 6H_2O(l) \rightarrow 4H_3PO_3(aq)$ Al_2O_3 is <u>insoluble</u> in water, hence water will remain at pH = 7. (Note: the description of “violent” or “vigorous” for reaction of P_4O_6 is necessary.)	[3]
	(b)	W has a giant ionic lattice structure. A lot of energy is required to overcome the strong electrostatic attraction between the cations and oxide ions, hence W has a high melting point. In the molten state, W can form free ions which act as mobile charge carriers to conduct electricity. X has a giant covalent structure. A lot of energy is required to overcome the strong and extensive covalent bonds between the atoms, hence X has a high melting point. In the molten state, X does not form mobile ions, suggesting that the bonding is not ionic. Y has a giant ionic lattice structure. A lot of energy is required to overcome the strong electrostatic attraction between the cations and oxide ions, hence Y has a high melting point. In the molten state, Y can form mobile ions to conduct electricity. Z has a simple covalent structure. Less energy is required to overcome the weaker intermolecular forces between molecules of Z, hence Z has a lower melting point / Since Z hydrolyses in water to form a strongly acidic solution, it is a non-metal oxide. In the molten state, Z does not form free ions to act as mobile charge carriers, suggesting that the bonding is not ionic. (Note: Cannot merely repeat the properties, e.g. high melting point and good electrical conductivity, without explaining the reasons for its high melting point and good electrical conductivity. It is recommended to use at least 2 properties in your explanation.)	[4]
	(c)	(i) $2MnO_4^- + 5SO_2 + 2H_2O \rightarrow 2Mn^{2+} + 5SO_4^{2-} + 4H^+$	
		(ii) The end-point is reached when the solution in the conical flask turns from colourless to pink. (Note: the solution does not turn from purple to colourless as MnO_4^- is added into the solution containing the dissolved SO_2 .)	
		(iii) Amount of $MnO_4^- = \frac{20.5}{1000} \times 0.0200 = 4.10 \times 10^{-4}$ mol $2MnO_4^- \equiv 5 SO_2$ Amount of $SO_2 = \frac{5}{2} \times 4.10 \times 10^{-4} = 1.03 \times 10^{-3}$ mol	[5]

		Concentration of $\text{SO}_2 = 1.03 \times 10^{-3} \div \frac{25.0}{1000} = 0.0412 \text{ mol dm}^{-3}$	
	(d)	$\text{pH} = -\lg K_a + \lg \frac{[\text{salt}]}{[\text{acid}]}$ $= -\lg (6.3 \times 10^{-5}) + \lg \frac{0.03}{0.01}$ $= 4.68$	[2]
	(e)	Benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$) undergoes hydrolysis most easily, followed by chloromethylbenzene ($\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$), followed by chlorobenzene ($\text{C}_6\text{H}_5\text{Cl}$) which undergoes hydrolysis the least easily. In benzoyl chloride, the carbonyl carbon in the $-\text{COCl}$ group is <u>highly electron-deficient</u> as it is bonded to <u>2 electronegative elements</u>, oxygen and chlorine. Hence it is most easily attacked by H_2O nucleophile in hydrolysis. In chloromethylbenzene, the carbon atom is <u>less electron-deficient</u> than that in benzoyl chloride as it is bonded to only <u>1 electronegative element</u>, chlorine. Hence it is less easily attacked by H_2O nucleophile. In chlorobenzene, p orbital of chlorine overlaps with the π electron cloud of the benzene ring. Hence, the lone pair of electrons on Cl atom can delocalise into the benzene ring, resulting in the C–Cl bond having a <u>partial double bond character</u>. Hence the C–Cl bond is stronger and does not break readily for nucleophilic substitution to occur.	[3]
	(f)	E:  F:  G: 	[3]
		[Total: 20]	

4	(a)	Enantiomerism. There are <u>2</u> enantiomers (due to 1 chiral carbon). (Note: C=C in six-membered ring will not exhibit cis-trans isomerism.)	[2]												
	(b)	 and $\text{O}=\text{C}=\text{O}$ and $\text{H}-\text{O}-\text{H}$	[2]												
	(c)		[2]												
	(d)	<table><tr><th>Information</th><th>Deduction</th></tr><tr><td>γ-terpinene undergoes <u>oxidation</u> to give H and J.</td><td>γ-terpinene has <u>C=C</u> / is an <u>alkene</u>.</td></tr><tr><td>Both H and J undergo <u>acid-carbonate reaction</u> with Na_2CO_3 to CO_2.</td><td>Both H and J has <u>-COOH group</u>.</td></tr><tr><td>Both H and J undergo <u>condensation</u> with 2,4-DNPH but <u>does not undergo oxidation</u> with Fehling's solution.</td><td>Both H and J are <u>ketones</u>.</td></tr><tr><td>H gives <u>positive iodoform test</u> but J does not.</td><td>H has a <u>-COCH₃ group</u> but J does not.</td></tr><tr><td>K undergoes <u>oxidation</u> with hot acidified $\text{Cr}_2\text{O}_7^{2-}$ to give J.</td><td>Primary and secondary alcohol in K oxidised to <u>carboxylic acid</u> and <u>ketone</u> group in J respectively.</td></tr></table>	Information	Deduction	γ -terpinene undergoes <u>oxidation</u> to give H and J .	γ -terpinene has <u>C=C</u> / is an <u>alkene</u> .	Both H and J undergo <u>acid-carbonate reaction</u> with Na_2CO_3 to CO_2 .	Both H and J has <u>-COOH group</u> .	Both H and J undergo <u>condensation</u> with 2,4-DNPH but <u>does not undergo oxidation</u> with Fehling's solution.	Both H and J are <u>ketones</u> .	H gives <u>positive iodoform test</u> but J does not.	H has a <u>-COCH₃ group</u> but J does not.	K undergoes <u>oxidation</u> with hot acidified $\text{Cr}_2\text{O}_7^{2-}$ to give J .	Primary and secondary alcohol in K oxidised to <u>carboxylic acid</u> and <u>ketone</u> group in J respectively.	
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		Structures:		
				 or 
		H	J	γ-terpinene
				[8]
(e)	(i)	$\text{C}_{10}\text{H}_{16}(\text{l}) + 14\text{O}_2(\text{g}) \rightarrow 10\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\text{l})$		
	(ii)	Enthalpy change of combustion of limonene $= 2 \text{ BE}(\text{C}=\text{C}) + 8 \text{ BE}(\text{C}-\text{C}) + 16 \text{ BE}(\text{C}-\text{H}) + 14 \text{ BE}(\text{O}=\text{O})$ $- 20 \text{ BE}(\text{C}=\text{O}) - 16 \text{ BE}(\text{O}-\text{H})$ $= 2(610) + 8(350) + 16(410) + 14(496) - 20(805) - 16(460)$ $= \underline{-5936 \text{ kJ mol}^{-1}}$ (or $-5940 \text{ kJ mol}^{-1}$ to 3 s.f.)		
	(iii)	Molar mass of limonene $= 10(12) + 16(1) = 136 \text{ g mol}^{-1}$ Enthalpy change of combustion of limonene $= \frac{-5936}{136}$ $= -43.6 \text{ kJ g}^{-1}$ The enthalpy change of combustion of limonene per gram is <u>slightly less negative than</u> (or around the same as) that of conventional diesel fuel. OR Slightly less heat is evolved per gram of limonene combusted than per gram of diesel fuel combusted.		
				[6]
				[Total: 20]

5	(a)	(i)	The covalent radius of silicon is <u>half the distance</u> between the centres of two adjacent Si atoms in a Si–Si covalent bond in the solid silicon lattice.	
		(ii)	Argon does not form compounds / share electrons. Hence, it exists as single atoms and does not have a covalent radius.	
		(iii)	Chlorine has <u>fewer protons in its nucleus</u> and hence a <u>smaller nuclear charge</u> than argon, while <u>shielding effect</u> is about the same. Hence, chlorine has a <u>smaller effective nuclear charge</u> . Thus, the <u>valence electrons of chlorine are less attracted by the positive nucleus</u> , causing chlorine to have a <u>larger van der Waals radius</u> than argon.	[3]
	(b)	(i)	 (Note: An O²⁻ ion forming a dative bonded to an oxygen atom is not accepted.)	
		(ii)	$\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$ (Note: Na₂O cannot be the product since it is an “alkaline solution”.)	[3]
	(c)	(i)	For the following rate equation, rate = $k[\text{A}]^m[\text{B}]^n$, the order of reaction is (m+n). OR The order of reaction is the sum of the powers of the <u>reactants' concentration term</u> in the experimentally-determined <u>rate equation</u>.	
		(ii)	Half-life is the <u>time taken</u> for the <u>concentration of a reactant</u> to be <u>halved</u> .	[2]

	(d)	(i)	 Since half-life is constant at 330s, order of reaction with respect to $[H_2O_2]$ is 1.	
		(ii)	 initial rate = gradient of tangent at $t = 0s$ $= \left \frac{0.0200 - 0.0030}{0 - 400} \right $ $= 4.25 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ (Note: Rate of a reaction is always positive.)	

		(iii) Let rate = $k[\text{H}_2\text{O}_2][\text{H}^+]^m[\text{I}^-]^n$ Keeping $[\text{H}_2\text{O}_2]$ and $[\text{I}^-]$ constant, $\frac{(\text{rate})_1}{(\text{rate})_2} = \left(\frac{[\text{H}^+]_1}{[\text{H}^+]_2}\right)^m$ $\frac{8.4 \times 10^{-5}}{6.3 \times 10^{-5}} = \left(\frac{0.400}{0.300}\right)^m$ $m = 1$ Hence, order of reaction with respect to $[\text{H}^+]$ is 1. Keeping $[\text{H}_2\text{O}_2]$ constant, $\frac{(\text{rate})_1}{(\text{rate})_3} = \left(\frac{[\text{H}^+]_1}{[\text{H}^+]_3}\right)^1 \left(\frac{[\text{I}^-]_1}{[\text{I}^-]_3}\right)^n$ $\frac{8.4 \times 10^{-5}}{2.1 \times 10^{-5}} = \left(\frac{0.400}{0.200}\right)^1 \left(\frac{0.200}{0.100}\right)^n$ $n = 1$ Hence, order of reaction with respect to $[\text{I}^-]$ is 1.	
		(iv) rate = $k[\text{H}_2\text{O}_2][\text{H}^+][\text{I}^-]$ $8.4 \times 10^{-5} = k(0.0200)(0.400)(0.200)$ $k = 0.0525 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$	[9]
	(e)	(i) LiAlH ₄ in dry ether / NaBH ₄ / H ₂ with Ni catalyst, heat	
		(ii) Reagents: K₂Cr₂O₇(aq), H₂SO₄(aq) To obtain propanal as the main product, <u>heat</u> the reaction mixture with <u>immediate distillation</u> to <u>prevent further oxidation of the propanal</u>.	[3]
		[Total: 20]	